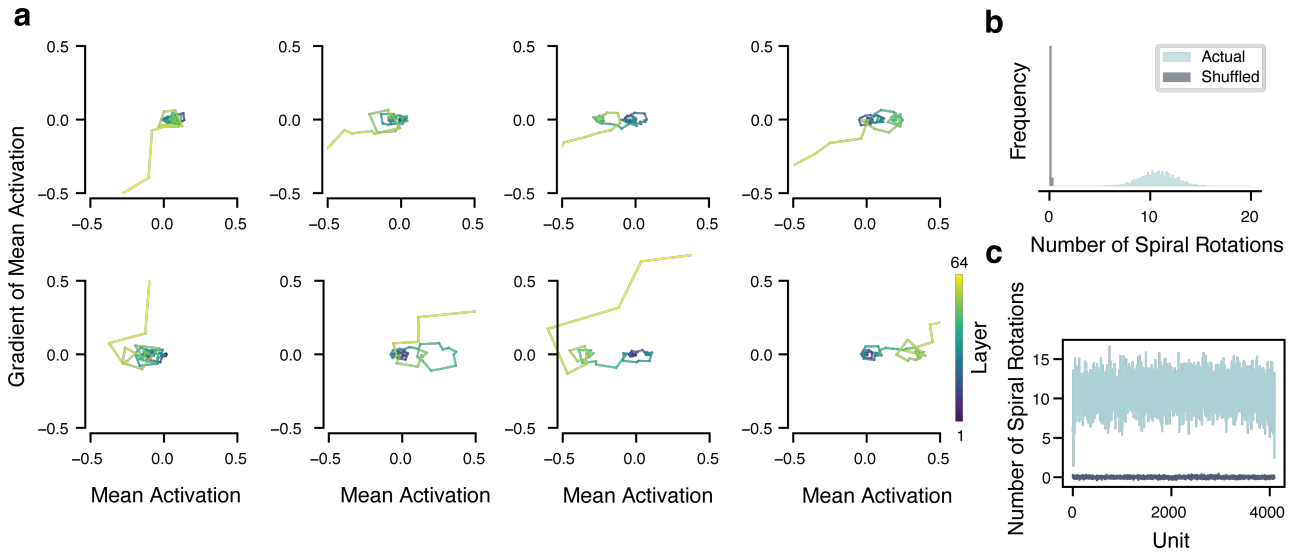


*Figure 2.* Portraits of Individual RS Units Show Rotational Dynamics akin to Unstable Periodic Orbits. **A:** Portraits of individual units in activation-gradient space, where the gradient is taken over the 64 effective sublayers. **B:** Distribution of the estimated number of rotations each unit performs in this phase space compared to a control in which the layer order was shuffled 1000 times for each unit. The mean number of rotations over the layers is 10.74 for the RS units and $\sim 0$ for the shuffle controls. **C:** The number of rotations for each for the 4096 units in the RS and their shuffle controls.

consecutive layers $l$ and $l+1$, we computed the mutual information using kernel density estimation:

$$\mathbf{MI}(l, l+1) = \sum p(x,y) \log \left( \frac{p(x,y)}{(p(x)p(y)} \right)$$

where $p(x,y)$ is the joint probability density of activations at layers $l$ and $l+1$, and $p(x)$ and $p(y)$ are their respective marginal densities.

We implemented this calculation using Gaussian kernel density estimation to handle the continuous nature of the activation space. For numerical stability, we added a small constant (1e-10) to avoid division by zero and taking logs of zero. The mutual information was computed in nats using the natural logarithm.

The computation was performed independently for each unit in the residual stream, using the distribution of activations derived from $N = 1000$ batches, to track how different components of the representation evolved through the layers.

### 2.6. Dynamics of Individual RS Units

To examine the dynamics of individual RS units' activations across layers, we created phase portraits in 2D phase space defined by each unit's activation value and the gradient of that activation across layers. For each unit $i$, we constructed a phase portrait where the x-axis represents the unit's activation $a_l^i$ at layer $l$, and the y-axis represents its gradient $\nabla a_l^i = \frac{d}{dl} a_l^i$.

The resulting portraits revealed rotational dynamics. To quantify these, the number of rotations in this 2D space was calculated by tracking the cumulative change in angle of the tangent vector along this trajectory. Specifically, for each unit we centered the trajectory at the origin by subtracting initial values:

$$x_l = a_l^i - a_0^i$$
$$y_l = \nabla a_l^i - \nabla a_0^i$$

We then computed tangent vectors between consecutive points:

$$\Delta x_l = x_{l+1} - x_l$$
$$\Delta y_l = y_{l+1} - y_l$$

Subsequently we calculated the angles of these tangent vectors:

$$\theta_l = \arctan 2(\Delta y_l, \Delta x_l)$$

Following this, we computed angle changes between consecutive points, adjusting for discontinuities at $\pm\pi$:

$$\Delta\theta_l = \theta_{l+1} - \theta_l$$

Finally, the total number of rotations was calculated as:

$$R = \frac{1}{2\pi} \sum_l \Delta\theta_l$$

To establish statistical significance, we compared the observed number of rotations with a null distribution generated

by randomly permuting the layer ordering 1000 times for each unit. This shuffle control preserved the distribution of activations while disrupting any systematic rotational structure.

### 2.7. Dimensionality Reduction with a Compressing Autoencoder

To analyze the high-dimensional activation patterns in the RS, we trained an autoencoder on the RS activations and visualized the trajectories across layers in reduced dimensional space. The compressing autoencoder[1] was trained to minimize reconstruction error while learning a low-dimensional representation of activation patterns. The architecture consists of an encoder and decoder, each with $k$ layers where $k$ is determined by the input dimension $d_{in} = 4096$ and target bottleneck dimension $d_{bottle} = 2$. The dimensions of intermediate layers follow a geometric progression, with each layer $i$ having dimension:

$$d_i = d_{in} \cdot r^i$$

where $r = (d_{bottle}/d_{in})^{1/(k-1)}$ is the reduction ratio between consecutive layers.

The Wikitext dataset was used for training and evaluating the CAE, with a train set of 85k batches and test set of 15k batches, each of which contained 64 samples (i.e. an RS vector for each layer).

Each layer consists of a linear transformation followed by layer normalization and ReLU activation (except for the final encoder and decoder layers which omit these nonlinearities). The model was trained using the Adam optimizer with learning rate $\alpha = 10^{-3}$ to minimize the mean squared error loss:

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{x}_i - f_\theta(\mathbf{x}_i)\|_2^2$$

where $\mathbf{x}_i$ are the input activation patterns and $f_\theta$ is the autoencoder with parameters $\theta$. Training proceeded for a maximum of 100 epochs with early stopping based on validation loss with a patience of 10 epochs. The model achieving the lowest validation loss was retained.

### 2.8. PCA and Perturbation of Activation Trajectories

To better understand the trajectories of the RS in reduced dimensional space and perform interpretable perturbations in these trajectories in reduced space, we performed Principal Component Analysis (PCA) using Singular Value Decomposition (SVD). For a dataset of $N$ samples with activations from $L$ layers, each of dimension $D$, we first reshape the activation tensor $\mathbf{RS} \in \mathbb{R}^{N \times 2L \times D}$ into a matrix $\mathbf{X} \in \mathbb{R}^{2NL \times D}$ by combining the sample and layer dimensions. Then we center the data by subtracting the mean: $\mathbf{X}_c = \mathbf{X} - \boldsymbol{\mu}$, where

$$\boldsymbol{\mu} = \frac{1}{2NL} \sum_{i=1}^{2NL} \mathbf{x}_i$$

Then we compute the SVD of the centered data:

$$\mathbf{X}_c = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$$

where $\mathbf{U} \in \mathbb{R}^{2NL \times D}$, $\boldsymbol{\Sigma}_{\mathbf{diag}} \in \mathbb{R}^D$, and $\mathbf{V} \in \mathbb{R}^{D \times D}$

For subsequent analysis, we project the data onto the first two principal components:

$$\mathbf{Z} = \mathbf{X}_c \mathbf{V}[:, :2]$$

where $\mathbf{V}[:, :2]$ contains the first two right singular vectors. The explained variance ratio for component $k$ is computed as:

$$r_k = \frac{\sigma_k^2}{\sum_{i=1}^{d} \sigma_i^2}$$

where $\sigma_k$ is the $k$-th singular value.

The resulting low-dimensional representation $\mathbf{Z} \in \mathbb{R}^{(2NL) \times 2}$ is then reshaped to $\mathbb{R}^{N \times 2L \times 2}$ for visualization and analysis of activation trajectories.

To investigate how perturbations in the learned low-dimensional space affect model behavior, we systematically explored the 2D PCA space by creating a uniform grid of points to which we could teleport the activations at various stages (layers) in the RS trajectory. Specifically, we generated $n \times n$ evenly spaced points across a range $[r_{min}, r_{max}]$ in each principal component dimension, where $n$ is the number of points per dimension (typically 10) and $r$ represents the range of perturbation magnitudes. For each point $\mathbf{z}_i$ in this 2D grid, we projected it back to the original activation space using the inverse PCA transformation:

$$\mathbf{x}_i = \mathbf{z}_i \mathbf{V}[:2]^T + \boldsymbol{\mu}$$

where $\mathbf{V}[:2]$ contains the first two principal components and $\boldsymbol{\mu}$ is the mean of the original activation distribution. We then injected these reconstructed activations into specific layers of the language model by replacing the original activations at the input to the attention layer. This process was repeated across multiple network layers to analyze how perturbations at different depths affect the model's internal representations. As perturbations above the first layer we used a standard input prompt ("I'm sorry, Dave. I'm afraid I can't do that.") to record the initial trajectories. This input also served as a control to establish a baseline for comparison of perturbed trajectories.

[1] We term this a 'compressing' autoencoder (CAE) to distinguish from Sparse Autoencoders (SAEs) in the interpretability literature.

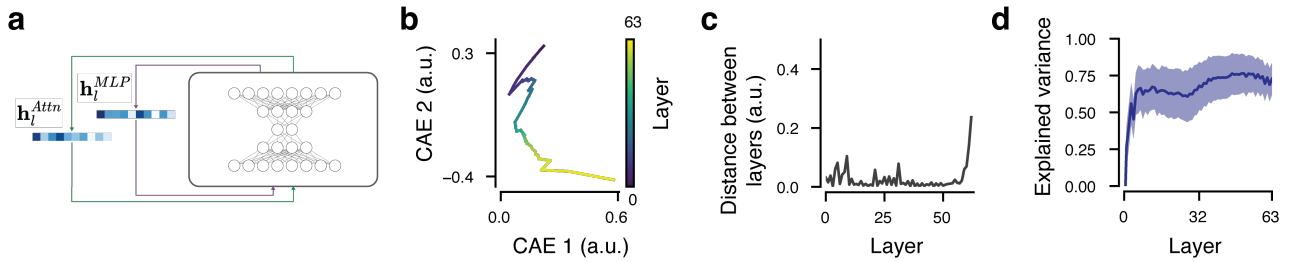


*Figure 3.* Compressing Autoencoder (CAE) Shows Dynamics of the RS in Reduced Dimensional Space **A:** The CAE was trained to pass RS vectors at individual pre-attention and pre-MLP sublayers through a bottleneck, and reconstruct the original vector. Results showing a CAE trained with 10 layers to reduce the dimensionality at the bottleneck to 2. **B:** Mean trajectory across $n = 1000$ test data samples. **C:** Distance in the reduced space between subsequent layers. **D:** Explained variance on the test set as a function of the layers.

## 3. Results

### 3.1. Transformer residual stream (RS) activations grow dense and are highly correlated over the layers

Our initial investigation into the activations of the RS showed that activations at the last token position for input sequences increase in magnitude over the layers (Fig. 1B). Most units showed low-magnitude activations at the lowest layers, with the majority increasing in magnitude progressively over the layers. Sorting the mean activations across $N = 1000$ data batches by the mean activation at the last layer, $\pi = \text{argsort}(\bar{\mathbf{h}}^{2L})$, revealed that activations not only grow dense as the layers progress, but that units tend to preserve their sign over the layers.

To quantify the continuity of representations between layers, we analyzed pairwise correlations between layer activations. We analyzed two types of transitions (Fig. 1C): within-layer transitions ($\mathbf{h}_l^{Attn} \rightarrow \mathbf{h}_l^{MLP}$) and cross-layer transitions ($\mathbf{h}_l^{MLP} \rightarrow \mathbf{h}_{l+1}^{Attn}$).

The within-layer correlations were consistently higher than the cross-layer correlations, suggesting different information processing regimes of attention and MLP operations, with the former producing smaller changes to the RS vectors. The correlation strength increased over layers for both transition types, with correlations starting high ($r > 0.8$) even in the earliest layers. For each unit, the distribution of correlations over the 63 layer transitions was binned into intervals $[0, 1]$ to create a density plot (Fig. 1D), revealing that despite the RS being a nonprivileged basis, most units maintain strong correlations throughout the network.

While correlations of activations revealed that individual units exhibited ever stronger linear relationships for successive layers, we were keen to examine the changes of the RS vector as a whole. Cosine similarity of layerwise RS vector pairs increased as a function of layers, with within-layer ($\mathbf{h}_l^{Attn} \rightarrow \mathbf{h}_l^{MLP}$) transitions more similar than cross-layer (Fig. 1E).

To characterize the dynamics of representation change through layers, we computed the velocity of the RS representation. The velocity profile showed a distinct pattern of acceleration through the model (Fig. 1F). Early layers maintain relatively constant velocities, while later layers exhibited a slight increase in velocity, with the steepest acceleration occurring in the last third of the model. This acceleration pattern holds for both within-layer and cross-layer transitions, though cross-layer transitions consistently showed higher velocities. This progressive acceleration of representational change, combined with our observations of increasing activation magnitudes and correlations, suggests that the transformer RS systematically amplifies certain representational directions in later layers.

Analysis of mutual information (MI) for given RS units at successive layers revealed three distinct phenomena in information flow (Fig. 1G,H). First, MI showed a sharp decline in early layers, with the steepest drops occurring at cross-layer transitions ($\mathbf{h}_l^{MLP} \rightarrow \mathbf{h}_{l+1}^{Attn}$). Second, the reduction in MI occured simultaneously with increasing linear correlations between layers (Fig. 1C,D). Third, the MI decrease coincided with growing activation magnitudes through the layers (Fig. 1B).

The apparent paradox between decreasing MI and increasing correlations suggests a systematic transformation of the representation space. While correlation captures only linear relationships, MI measures both linear and nonlinear dependencies. This pattern, combined with our observation of decreasing dimensionality through the layers, suggests that the model may be redistributing information across more dimensions while favoring simpler, linearly-aligned features in later layers over complex nonlinear relationships.

### 3.2. RS Unit Phase Space Exhibits Rotational Dynamics

For each unit $i$, we constructed phase portraits by plotting the unit's activation value $a_l^i$ against its gradient $\nabla a_l^i = \frac{d}{dl} a_l^i$ across layers. Analysis of individual RS units in activation-gradient space revealed rotational dynamics characteristic of unstable periodic orbits (Fig. 2A). While most trajectories